



Neighborhood interactions and local edaphic gradients have a weak influence on long-term vegetation dynamics in an old-growth forest community

Stephen J. Murphy^{a,*}, Brian C. McCarthy^b

^a Department of Evolution, Ecology, and Organismal Biology, 318 W 12th Ave, The Ohio State University, Columbus, OH 43210, United States

^b Department of Environmental and Plant Biology, 315 Porter Hall, Ohio University, Athens, OH 45701, United States



ARTICLE INFO

Article history:

Received 23 September 2016

Accepted 28 December 2016

Available online 16 January 2017

Keywords:

Dysart woods

Mesophication

Mortality

Neighborhood model

Oak decline

Recruitment

Succession

ABSTRACT

The Eastern Deciduous Forest (EDF) region is undergoing a shift in species composition, the most dramatic trend being an increase in mesic taxa such as maple (*Acer* spp.) and American beech (*Fagus grandifolia*), and a subsequent decline in xeric species such as oaks (*Quercus* spp.). While several hypotheses have been proposed to explain these trends, most focus on allogenic processes such as disturbance and climate change, and little attention has been given to local species interactions. To help bridge this gap, we first quantify long-term changes in community composition of an old-growth forest using census data collected over a fifteen-year interval. Within two 50 × 70 m plots, all stems ≥ 2.5 cm DBH were tagged and measured in 1996, and their spatial coordinates were mapped. Seedlings were measured in smaller subplots, and soil cores were sampled at these locations. Hemispherical photographs were then taken to quantify the light environment, and these procedures were replicated in 2011. As expected, we find an increase in the dominance of both American beech and sugar maple (*Acer saccharum*), and a marked decline for most others, particularly white oak (*Quercus alba*). Neighborhood models incorporating the density, basal area, and composition (i.e., conspecific versus heterospecific) of stems surrounding focal individual trees reveal that local density-dependent interactions were indeed weak for both American beech and sugar maple. At the same time, however, all other species experienced similarly weak effects. Likewise, soil resource gradients had little influence on long-term demographics, contrary to our expectations. Together, our results indicate that the changes observed across the Eastern Deciduous Forest region may indeed be most strongly impacted by allogenic disturbance processes, although we still encourage further research on the potential influence of local species interactions at mediating these dynamics, and expect neighborhood models to become an important tool in understanding the causes of compositional shifts in the EDF region.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Forest community dynamics are driven by both autogenic interactions such as competition for soil resources, and allogenic processes like fire, climate change, and herbivory (Tansley, 1935). Despite decades of research on this topic, it remains unclear how the relative contributions of each of these processes impact the long-term dynamics of forest populations. Moreover, we know little about how such information can be used to make predictions about the future. Understanding the roles of both internal and external ecological processes is critical for understanding why

forests change, and for developing appropriate management strategies (Oliver and Larson, 1996). For example, if local competition is most important, then stand thinning or planting strategies that reduce conspecific density would be required. Alternatively, if fire or herbivore outbreaks are more important, then management strategies that specifically target the dynamics of these disturbance regimes would be most effective. It is also likely that these two types of processes interact in tandem to drive vegetation dynamics in forests ecosystems.

For the case of within-stand dynamics, our ability to quantify tree-to-tree interactions has been greatly enhanced in recent decades through a shift from a whole-community or ecosystem approach, to an individualistic perspective where data are retained at the stem-level (Liu and Ashton, 1995). These so-called neighborhood analyses allow for a more detailed understanding of how

* Corresponding author.

E-mail addresses: murphy.1132@osu.edu (S.J. Murphy), mccarthy@ohio.edu (B.C. McCarthy).

individuals influence community dynamics, and for understanding the importance of intraspecific demographic variation. The explicit incorporation of space into models of species interactions has also been helpful because it allows for the specification of biotic and abiotic neighborhoods that are ecologically relevant (Legendre and Fortin, 1989; Liebholt and Gurevitch, 2002). For example, the influence of density, composition, nearest distance to neighboring tree, and size, on the growth of a target individual can be estimated (Canham et al., 2004; Zhao et al., 2007). Alternatively, local spatial variation in soil resource gradients can be quantified, and the response of individual trees to important nutrients can be determined (Baribault et al., 2010).

Neighborhood analyses have also shed light on how the taxonomic identity of proximal neighbors influences demography (Hubbell et al., 2001; Zhao et al., 2004; Wiegand et al., 2009). An important and oft-cited idea in coexistence theory posits that an increase in the density, proximity, and size of conspecific neighbors leads to a reduction in the performance of focal individuals (Janzen, 1970; Connell, 1971). This may be due to an increased abundance of specialist herbivores and/or pathogens as conspecific neighbors increases (Bagchi et al., 2014). In turn, such patterns are expected to result in a compensatory effect on survival for species at low relative abundances, which could be an important stabilizing mechanism (Connell et al., 1984). While the vast majority of work on this subject has been conducted in species rich tropical rainforests, recent work has shown that the mechanism is likely pervasive across the Eastern Deciduous Forest (EDF) of the United States. (Johnson et al., 2012). Ultimately, an increase in the strength of density dependence (e.g., through a population increase of the natural enemy of a species) should result in a reduction in relative abundance for the target tree species. Thus, understanding the role of density-dependent interactions is essential for understanding long-term community changes, including those occurring in the EDF region.

A dramatic and widespread example of forest composition change in the eastern U.S. is a shift in dominance from xerophytic to mesophytic communities (Dyer, 2006; Nowacki and Abrams, 2008; McEwan et al., 2011). Specifically, a reduction in the survival and regeneration of oaks (*Quercus* spp.) and hickories (*Carya* spp.) has been observed with a concurrent increase in regeneration of species such as maple (*Acer* spp.) and American beech (*Fagus grandifolia*) (McEwan et al., 2011). These compositional shifts have been hypothesized to lead to dramatic changes in the structure and function of these communities, as well as to economic consequences due to declines of valuable timber species (McShea et al., 2007). However, the root cause of these changes remains unknown. A review outlining the possible mechanisms influencing mesophication focused solely on external (i.e., allogenic) drivers (McEwan et al., 2011). The authors outline several possible mechanisms, including a reduction in natural fire frequency, white-tailed deer (*Odocoileus virginianus*) herbivory, human land use, and climate change, among others. While these hypotheses are indeed likely important, ecologists have been unsuccessful at linking them directly to the observed changes described above. In contrast, the role of internal interactions such as neighborhood competition and soil resource acquisition to mesophication dynamics in the EDF has received no attention to our knowledge within this context.

In this study, the role of neighborhood competition and response to local soil conditions on forest dynamics was addressed using long-term data collected from Dysart Woods (Belmont County, Ohio), an old-growth remnant tract that is experiencing a dramatic transition from white-oak (*Quercus alba*) to beech-maple dominance (Lafer and Wistendahl, 1970; McCarthy et al., 2001). We first determine the mortality and recruitment rates of all tree species over a fifteen-year interval at the site. We then

determine how the local abiotic and biotic neighborhood influences the growth and survival of various species. We hypothesize that American beech and maple are experiencing weaker density dependence relative to other species within the community. Likewise, we expect that these species have benefited from changes in soil resources over this time period, while other species have been more negatively impacted. Finally, we study the spatial patterning of recruitment over the census interval to determine the effects of dispersal limitation and competition on plant recruitment. As above, we hypothesize that negative density-dependent recruitment is weaker for both American beech and maple than for all other species in the Dysart Woods community.

2. Materials and methods

2.1. Study site

Like most of the old-growth communities in the Eastern Deciduous Forest (EDF) today, Dysart Woods is a small remnant tract (~23 ha) surrounded by secondary woods, old-fields, and pastures. The area lies within the unglaciated and hilly southeastern portion of the state of Ohio. The old-growth community is situated within a larger 250 ha field lab that is used primarily for research, farming, and American chestnut (*Castanea dentata*) restoration. The old-growth portion is divided into two approximately equal sized stands that are separated by an unpaved county road. The climate of the region is classified as temperate continental, having a mean winter and summer temperature of -2.2 and 20.0 °C, respectively, and an annual precipitation rate of approximately 1000 mm (Rubel et al., 1981).

2.2. Field methods

2.2.1. Trees

In 1996, two permanent plots were established at Dysart Woods to document and monitor the vegetation (McCarthy et al., 2001). Both plots, one in each stand of old-growth forest, are 50×70 m in dimension and were permanently marked using steel rebar. Plots were then divided into 35, 10×10 m subplots. Within each subplot, all stems ≥ 2.5 cm diameter at breast height (DBH) were tagged, measured, identified to species, and plotted on an X-Y Cartesian coordinate system. In 2011, the same procedures were repeated. The status (i.e., alive or dead) of each of the stems from the 1996 sample was recorded in 2011, and trees still alive were re-measured. Furthermore, all new stems ≥ 1 cm DBH were measured and mapped to study recruitment dynamics over the fifteen-year period. In both sampling years, an assessment of the overall health and vigor of each tree was taken (McLaughlin et al., 1992; McCarthy et al., 2001). Decline was assessed based off visual observations of branch dieback, crown transparency, leaf discoloration, and epicormic sprouting. These data were then used to calculate a decline score on a scale from zero (completely healthy) to one hundred (dead) (McCarthy et al., 2001).

2.2.2. Seedlings

The woody seedling understory was sampled using circular subplots placed at the centers of each of the $70 \times 10 \times 10$ m quadrats described above. Subplots 2-m^2 and 5-m^2 in area were used to census two size classes of seedlings. Class-I seedlings were defined as individuals ≥ 5 cm in height and < 30 cm tall, and were sampled within the smaller subplot. Class-II seedlings were those individuals ≥ 30 cm in height and < 2.5 cm DBH, and were sampled in the larger subplots. Relative abundances of seedlings in 2011 were then compared to the 1996 census.

2.2.3. Environmental data

The abiotic environment at Dysart Woods was quantified by measuring a suite of soil properties, as well as overstory solar irradiance (McCarthy, 1997; DeForest and McCarthy, 2011). Soil samples were collected at the outer perimeters of each of the 70 5-m² circular subplots where seedlings were sampled. Three individual soil cores were taken at a depth of 10 cm in each subplot using a small garden trowel and combining them into a single bulk sample (McCarthy, 1997; McCarthy et al., 2001). Field fresh soil was then transported to the laboratory, where percent moisture was measured gravimetrically and nitrate was analyzed using the cadmium reduction method. Remaining soils were then air-dried and passed through a 2 mm sieve. Percent organic matter was measured as loss at ignition at 550 °C. pH was determined using a 1:1 soil:water ratio on a Corning pH meter. Base cations (Mg, Ca, K, P and Al) were determined using a Mehlich III extraction procedure (Mehlich, 1984) and a Varian 20 Atomic Absorption mass spectrometer (Varian Inc., Walnut Creek, CA). The light environment was measured using hemispherical photography. Photographs were taken at the corners of each 10 × 10 m subplot on a uniformly overcast day, for a total of 96 samples across both plots, which were then analyzed using the Gap Light Analyzer program (Frazer et al., 1999). All measurements taken in 1997 were replicated in 2011 except for texture, which is unlikely to change over the timescale of this study. For base cations and aluminum, the original atomic absorption mass spectrometer used in the first analysis was no longer available, and a Varian 700-ES inductively coupled optical emission spectrometer (ICP-OES) was used instead.

2.2.4. Plant neighborhoods

The surrounding biotic neighborhood of trees at Dysart Woods was calculated within a 5-m radius from each tree. Each individual was visited and the total stem densities and basal areas of both conspecific and heterospecific individuals ≥2.5 cm DBH were calculated. Edge correction methods were used to estimate full neighborhoods of individuals that were located within 5 m of the plot

boundaries (Wiegand and Moloney, 2004). A 5-m radius was chosen because previous studies have shown neighborhood influences to be most intense at this scale (Hubbell et al., 2001), and because we wanted to reduce the amount of edge-correction area necessary to calculate neighborhoods of all trees sampled. Due to possible biases in the results related to the size of neighborhood radii, we also reran all analyses using a neighborhood radius of 7.5 and 10 m. These analyses indicated that there were no biases in our results or conclusions based on the choice of the neighborhood radius used (see Figs. S1 and S2).

2.3. Statistical analyses

2.3.1. Stand structural attributes

Relative basal area (RBA) and relative density (RDEN) were used to summarize the abundances of all species sampled. An overall relative importance value (RIV) was then calculated for each species by taking the average of RBA and RDEN for each species. For seedlings, only density was used as a measure of abundance. Relative cumulative frequency distributions (RCFD) were then constructed for the nine most common species at Dysart Woods for each of the two sampling periods. Finally, mortality and recruitment rates were calculated for the eight most common species using the equations from Sheil et al. (1995).

2.3.2. Growth and mortality models

The influence of soil resources and biotic neighborhood on plant growth was determined using multiple linear regression. For mortality, a binary error distribution was used (i.e., logistic regression) and regression coefficients were reported as odds ratios. Predictor variables used in the model included ln tree size (i.e., DBH), ln decline index score (Section 2.2.1), four plant neighborhood indices (Section 2.2.4), and soil properties (Section 2.2.3). The high dimensionality of the soils data was reduced using principal components analysis (John et al., 2007). The first two principal components were used as predictor variables in the growth and mortality

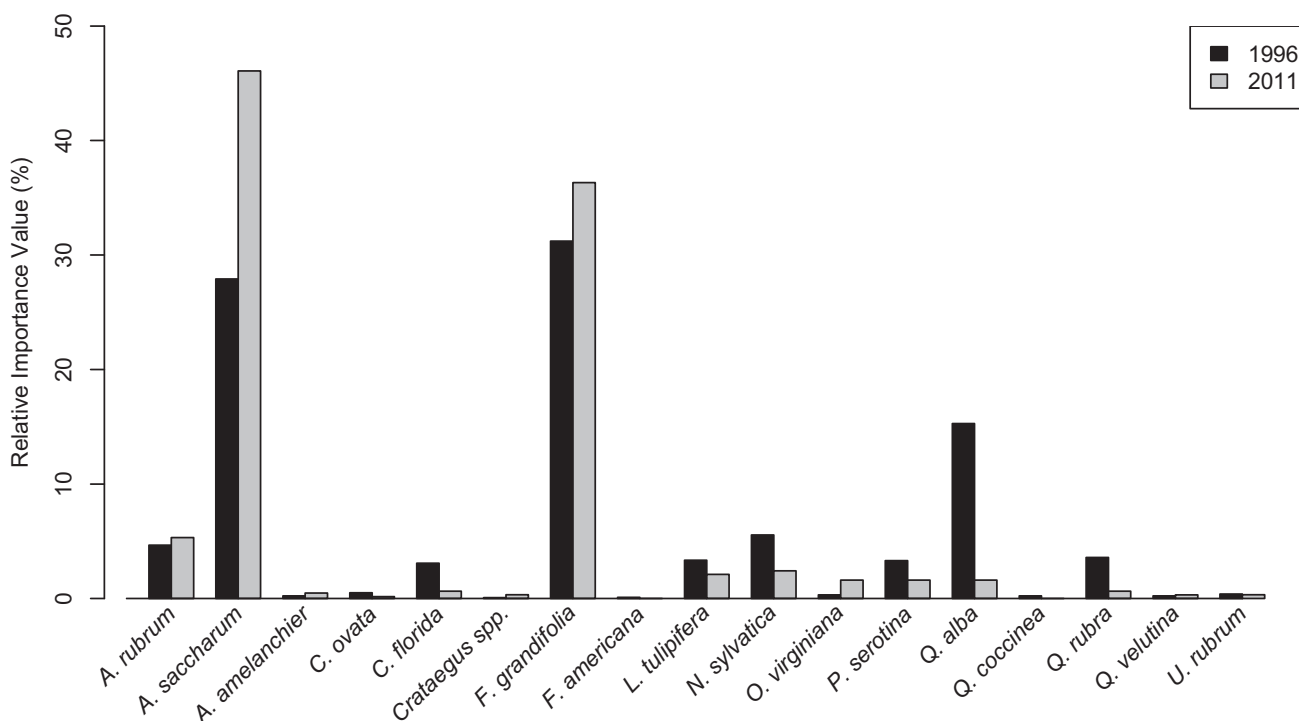


Fig. 1. Relative importance values (RIV) for all species in both plots at Dysart Woods in 1996 (black bars) and 2011 (gray bars). Species are listed left to right in alphabetical order.

models. All models were run for all species pooled, and for American beech and sugar maple (*Acer saccharum*) separately. Due to sample size restrictions, all remaining species were pooled into the category “other.” This classification was also biologically convenient, since American beech and sugar maple are the two species increasing rapidly in abundance, while nearly all others are declining.

2.3.3. Spatial autocorrelation

Spatial autocorrelation is a common problem in neighborhood models because individual trees likely do not represent independent sampling units (Hubbell et al., 2001). Thus, to test for the possible influence of spatial autocorrelation, we plotted the semivariances of the residuals for all model runs in our analysis. This was done by calculating the variance between observations (i.e., model residuals) for paired distances up to 20 m. If model residuals were significantly spatially autocorrelated, we would expect to see obvious patterns in these figures. In the end, our results indicated that spatial autocorrelation was not overtly present in any of our models (Fig. S3).

2.3.4. Spatial recruitment model

Lastly, we measured the spatial pattern of recruitment for American beech and sugar maple. We used the bivariate O-ring statistic between adults and new recruits (Wiegand and Moloney, 2014). The O-ring statistic quantifies the average expected number of newly recruited saplings as a function of distance from adult conspecific neighbors. To test whether the observed patterns deviate from a null expectation, Monte Carlo simulations were used to construct approximate 95% confidence envelopes. Adult locations were held constant, while the spatial locations of new recruits were randomized according to a heterogeneous Poisson process derived from the first order spatial pattern of adults. Simulations were run a total of 199 times, and the fifth highest and lowest realizations were used as the upper and lower bounds. This was done to control for the effects of dispersal limitation, allowing us to test for fine-scale dispersion or aggregation between adults and recruits. This method allowed us to specifically address whether recruitment was hindered by proximity to larger conspecific adults or not.

3. Results

3.1. Demographic rates

Patterns of mortality and recruitment among species were highly consistent with oak decline and mesophication. Sugar maple and American beech both increased in relative importance over the study interval, while white and red oak declined sharply (Fig. 1). White oak (*Quercus alba*) fell from a relative importance value (RIV) of 15.3 in 1996 to just 1.6 in 2011, a 90% reduction over just 15 years. This was mostly due to a moderate level of mortality with no subsequent regeneration, coupled with the large increase in the density of other species. Of the 22 total living oak stems in 1996, only 15 (68%) survived until 2011. The increase in RIV for American beech and sugar maple was due primarily to increases in stem density. In fact, relative basal area of American beech decreased from 1996 to 2011 due to the death of two large individuals (DBH = 70.8 and 78.7 cm). Of all new recruits measured, 94% were American beech or sugar maple. Other noticeable changes in abundance were a large decrease in flowering dogwood (*Cornus florida*) and black cherry (*Prunus serotina*), and a large increase in the density of red maple (*Acer rubrum*) on the south plot.

Similarly, annual rates of mortality and recruitment were highly varied among species, as well as between the north and south plots

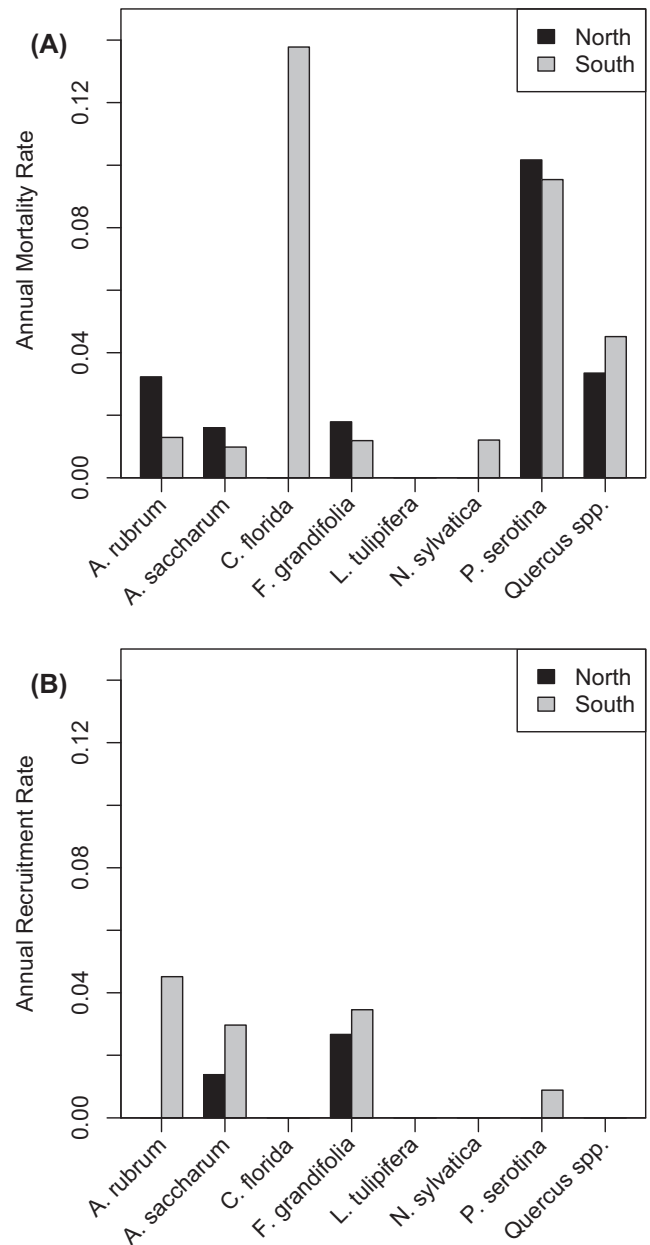


Fig. 2. Annual mortality (A) and recruitment (B) rates for the eight most common species at Dysart Woods. Rates in the north and south plots are indicated separately by black and gray bars, respectively.

(Fig. 2). Flowering dogwood and black cherry both had very high rates of mortality. Both sugar maple and American beech exhibited higher annual rates of recruitment than mortality, indicating that these species will likely continue to increase in abundance in the future. In contrast, all oak species showed high mortality rates on both plots, and not a single stem grew into the census over the 15-year study interval. These patterns are also reflected in the relative cumulative frequency distributions (Fig. S4).

The overall density of tree seedlings and shrubs recorded decreased over the study interval from a total of 174,857 Class-I and Class-II seedlings ha^{-1} in 1996, to 118,286 ha^{-1} in 2011 (Table S2). A significant portion of this decrease was due to a large reduction in the abundance of *Parthenocissus quinquefolia*, with ~83,000 stems ha^{-1} recorded in 1996 and just ~36,000 ha^{-1} recorded in 2011. Surprisingly, both American beech and sugar maple seedlings also decreased in abundance from 1996 to 2011,

despite the large increase in RIV for adults. Species that showed large increases in density included American ash (*Fraxinus americana*), shagbark hickory (*Carya ovata*), grapevine (*Vitis* spp.), and sassafras (*Sassafras albidum*).

3.2. Environmental data

Soil pH was greater in 2011 than in 1997 (Table 1). This increase was found on both plots, but more so on the north. There was no consistent pattern in changing soil fertility over the study interval, as measured by concentrations of base cations observed between sampling periods. While some base cations were significantly lower in 2011, others increased in abundance or showed no change (see Table 1 for tests). Overall, differences in soil properties between the north and south slopes were similar in 2011 as they were in 1997. Finally, average overstory light irradiance declined from 1996 to 2011, likely due to the increase in sapling density over the study period (Table 1).

3.3. Growth and mortality

Tree survival was strongly dependent on both size and decline score, with larger and healthier trees having a higher probability of survival (Fig. 3). The fitted logistic function indicated that the probability of survival based on DBH alone ranged from ~60% for the smallest trees to ~90% for the largest (Fig. 3). For the decline score, individuals with scores close to zero (i.e., healthy individuals) have a near 100% probability of survival, while individuals with the highest scores were as low as 20% (Fig. 3).

The influence of the biotic neighborhood and variation in soil gradients varied among species groups (Fig. 4). Tree growth was largely independent of the biotic neighborhood, except for American beech, which experienced a slight positive conspecific density-dependent effect on growth. Growth also seemed largely independent of the two soil PCA axes, except for a slight effect of PCA axis 2 on maple. Survival was similarly insensitive to neighborhood composition and soils. Except for a small increase in survival at higher

conspecific densities when all species were pooled, there were no observed effects of neighborhood composition, density, or basal on survival probability. Likewise, we observed only a small negative effect of soil PCA axis 1 on maple survival.

3.4. Spatial pattern of recruitment

Results from the bivariate O-ring models indicate that recruitment patterns in relation to canopy individuals were random when the intensity parameter of recruits was explicitly included (Fig. 5A–D). At all distances from adult conspecifics, recruitment was never significantly different from random, indicating that recruitment was independent of neighborhood composition.

4. Discussion

4.1. Long-term change

We found that Dysart Woods is currently experiencing a dramatic shift in species composition from oak to beech-maple dominance. While several large white-oak individuals still dominate the canopy, the seedling and sapling layers are devoid of this species, indicating that regeneration and replacement is highly unlikely. In addition to white oak, most species other than American beech and maple experienced declines in relative importance over the 15-year period, further indicating the likelihood of a future beech-maple dominated community. As mentioned previously, similar trends have been observed in other communities, and many competing hypotheses have been proposed to explain them. While there is some evidence that disruption of fire disturbance is to blame at Dysart Woods (McCarthy et al., 2001), the importance of fire in southeastern Ohio in the 18th and 19th century is much debated (Whitney, 1994). And while some suggest that historical forest fragmentation due to clear-cut logging may be to blame (McEwan et al., 2011), the fact that we observe these trends in a forest that has experienced minimal anthropogenic disturbance does not support this hypothesis.

Table 1
Summary of soil characteristics of Dysart Woods. 1997 data were obtained from McCarthy et al. (2001). The third column shows significance tests for changes in soil characteristics from 1997 to 2011. Two-sided *t*-tests were used when data were normally distributed, and the Kolmogorov-Smirnov test was used when they were not (indicated by asterisk).

	North 1997		North 2011		Significance tests	
	Mean	SD	Mean	SD	<i>t</i> / <i>W</i>	<i>P</i>
N-NO3	12.28	4.38	6.68	2.76	0.9021*	0.0045
P	6.28	1.94	6.4	2.15	−0.245	0.8
K	17.14	4.37	22.19	7.3	−3.48	<0.0010
Ca	81.77	136.45	72.97	28.65	0.514*	<0.0010
Mg	9.97	5.9	6.27	1.98	3.514	0.001
Al	156.77	18.42	75.99	13.38	20.334	<0.0001
pH	4.59	0.42	5.26	0.38	−7.151	<0.0001
%Moist	36.91	8.22	35.27	3.93	1.00*	<0.0001
% Global light	3.56	2.33	4.47	0.893	0.8091*	<0.0001
% O. matter	11.86	2.39	8.28	1.07	0.8286	<0.0001
	South 1997		South 2011		Significance tests	
	Mean	SD	Mean	SD	<i>t</i> / <i>D</i>	<i>P</i>
N-NO3	3.02	2.95	8.81	4.34	0.800*	<0.0001
P	1.79	1	2.55	0.99	−3.1024	0.0029
K	14.54	4.75	17.53	5.64	0.2571*	0.1975
Ca	42.86	108.38	48.19	45.63	0.5429*	<0.0001
Mg	6.63	6.69	5.91	4.55	0.2857*	0.1148
Al	165.34	22.27	83.61	13.45	0.9714*	<0.0001
pH	4.66	0.37	4.78	0.51	0.1714*	0.6828
%Moist	36.07	11.73	32.8	5.9	1.00*	<0.0001
% Global light	5.48	3.65	5.11	1.66	0.2292*	0.1607
% O. matter	12.37	2.64	9.11	1.85	0.5429*	<0.0001

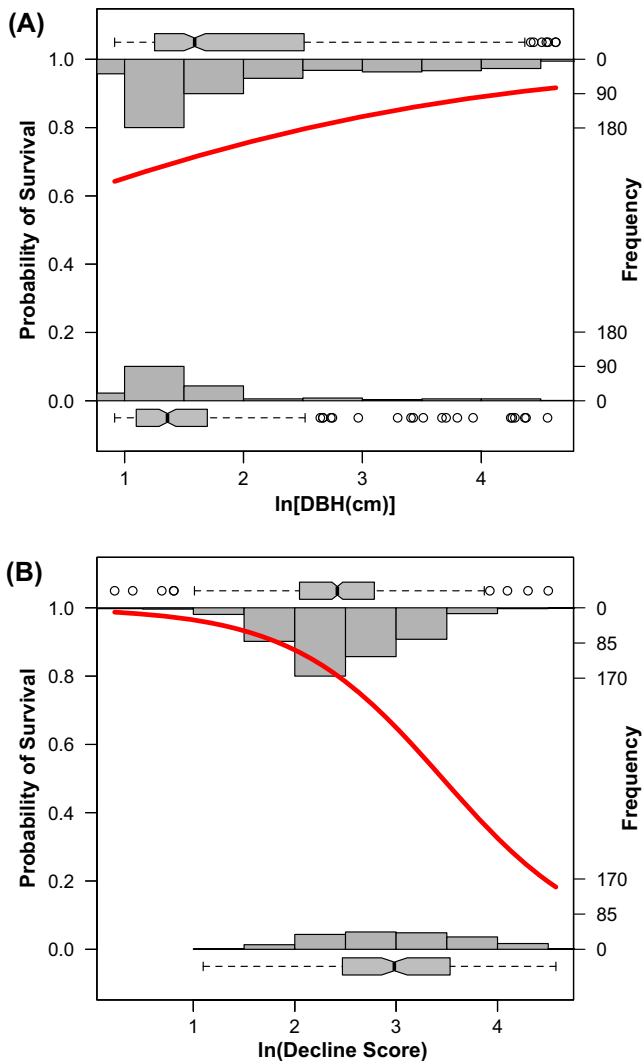


Fig. 3. Estimated logit probability of tree survival over fifteen years as a function of tree size (A) and initial health status (B). Histograms show the distribution of stems that survived and died along the x-axes.

4.2. Neighborhood effects on demography

While many hypotheses concerned with allogenic processes have received considerable attention in relation to oak decline and mesophication, little work has been done to assess the impacts of local competition on transitional dynamics in mixed oak communities. Richards and Hart (2011) documented gap dynamics in an Alabaman oak forest community, and found that recruitment in gaps was dominated by sugar maple and American beech. In contrast, they report that the probability of oak recruitment in these gaps was low except on the driest sites. Relatedly, Buchanan and Hart (2012) revealed through tree-ring analysis that endogenous canopy-gap disturbance dynamics necessary for oak recruitment may have become less frequent over time in the EDF. These kinds of localized demographic patterns are necessary for unraveling the complexity of mesophication trends, but have been understudied within this context.

We proposed a novel hypothesis, where American beech and maple experience weaker competitive effects than other species in the community, resulting in their observed increase in abundance. This was important to test because previous

work shows that differences in the strength of intraspecific neighborhood effects drive species abundance distributions (Comita et al., 2010; Kobe and Vriesendorp, 2011). Overall, our results provide support for our hypothesis that maple and American beech experience weak density-dependent control. However, the effects of neighborhood were similarly weak when analyzing other species, in contrast to our expectations. Growth over the fifteen-year interval was largely independent of neighborhood composition, except for a small positive effect of conspecific density on American beech growth. Similarly, survival seemed to be largely independent of biotic neighborhood, except for a small positive effect of conspecific density on survival when all species were grouped, which was likely due to positive habitat effects, and not to biotic facilitation *per se*. Together, our results indicate that differences in the strength of density-dependent interactions among species are unlikely to be contributing to long-term mesophication trends at our study site.

Our results generally lie in contrast to previous work that has found relatively strong support for negative density-dependent or Janzen-Connell type interactions among populations of forest trees (Comita et al., 2014). Most evidence for Janzen-Connell mediated mortality has been derived from studies focusing on tropical regions (Augspurger, 1984; Comita et al., 2014). However, there has been a steadily increasing body of work suggesting that these patterns are independent of latitude, and may be pervasive throughout forest communities globally (Lambers et al., 2002; McCarthy-Neumann and Kobe, 2010; Johnson et al., 2012). Thus, we expected to find some support for this hypothesis at our study site as well. Our lack of evidence for these same types of patterns at Dysart Woods can potentially be explained through several different lines of evidence. First, we expected to find very weak conspecific negative-density dependent interactions for both American beech and sugar maple. Because these two species make up 80% of the total number of individuals within the Dysart community, it may come as little surprise that we found no evidence for these patterns at the community level. In other words, sample sizes for other species, even when pooled, may have simply been too small to detect these trends, especially at the relatively small area analyzed here. Furthermore, by measuring the mortality of stems greater than 2.5 cm DBH, we are likely missing much of the strongest density-dependent interactions, which are known to occur at earlier life stages (Zhu et al., 2015). Considering these limitations, we recommend future studies seeking to identify underlying causes of mesophication dynamics to still consider the potential role of local species interactions.

4.3. Soil environment impacts

Our data indicate that the soil environment had little to no impact on the growth and survival of trees at Dysart Woods. For both growth and survival, maple was the only species to show any significant correlation with soil gradients, although the effect sizes were small in both cases. There is some evidence that soil resource gradients influence above ground net productivity (Baribault et al., 2010), but work on soil-dependent demography in trees is still generally lacking. Our results also did not support our expectation that soil acidity would decrease from the 1996 to 2011 census.

4.4. Other factors influencing growth and survival

Generally, the strongest and most consistent variables predicting survival probability and growth were the initial size of

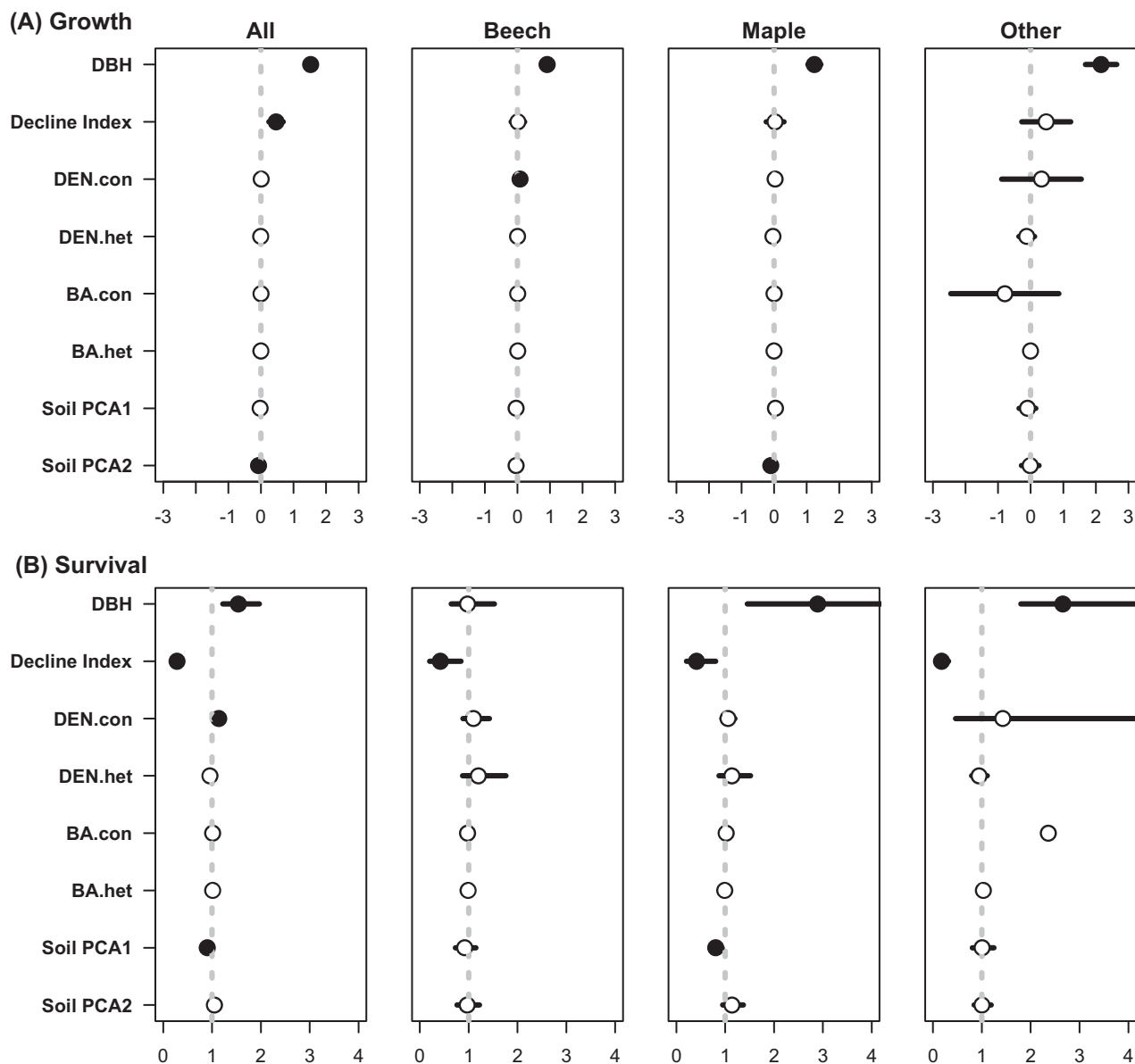


Fig. 4. Growth and survival models for trees at Dysart Woods over the 15-year study interval. Black dots indicate estimated model coefficients, and black bars indicate their standard error.

the target individual and its overall health status. These findings are in accordance with previous knowledge that survival probability increases with tree size, and excessive branch dieback, crown transparency, epicormic branching, and leaf discoloration are negatively associated with plant vigor (Barnes et al., 1998; Burns and Honkala, 1990). The decline index was also a better predictor of survival than for growth, and it was strongest for species other than American beech and sugar maple. This may be because most of the American beech and sugar maple individuals at Dysart were healthy, leading to little variation in the decline scores for these species. In contrast, many of the other species, particularly white oak, had low decline scores. And while the index used is somewhat subjective, our data show that such a method can accurately predict the demography of long-lived trees. The majority of forest modeling studies do not include any measure of initial tree health or vigor, which may be a particularly important piece of information to include in future monitoring efforts.

4.5. Recruitment dynamics

Overall, competitive influences on the regeneration and recruitment of American beech and sugar maple stems were not observed. If strong competitive interactions were important at regulating where new stems sprout and persist, we would have expected to see a significant overdispersed signal at small distance classes (i.e., <5 m). Therefore, our results indicate that the recruitment of American beech and sugar maple is largely free from density-dependence, which is likely contributing to their increase in dominance. The underlying causes for these dynamics is likely due to the high degree of shade tolerance of American beech and sugar maple. And while it would be interesting to know how recruitment dynamics are regulated for all other species at Dysart Woods, the predominance of American beech and sugar maple in this stratum precluded us from having a large enough sample size to do so. Therefore, it remains unclear whether species other than American beech and sugar maple experience stronger density-dependent recruitment effects.

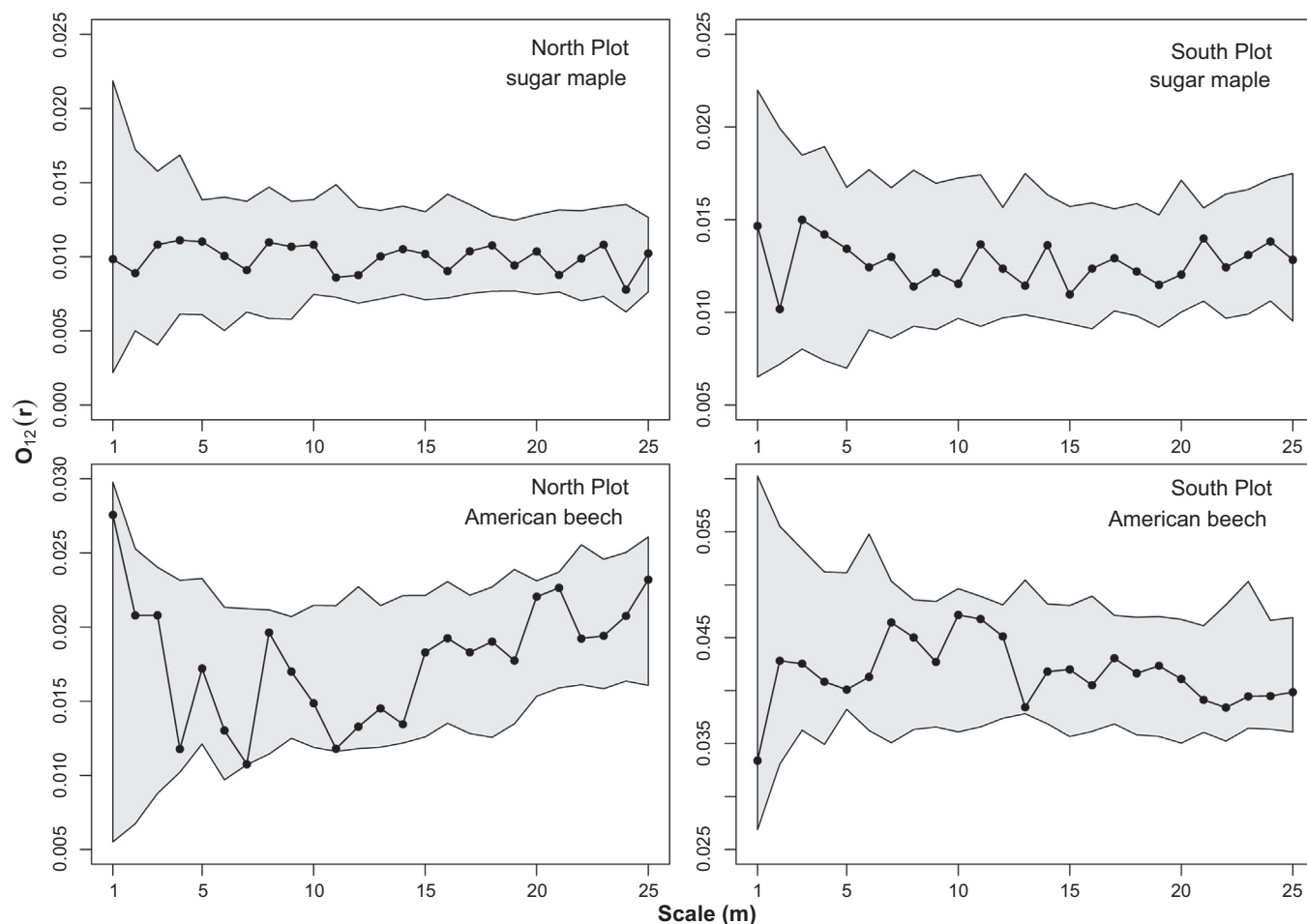


Fig. 5. Patterns of newly recruited stems (≥ 1 cm DBH in 2011 and not sampled in 1996) in relation to adult individuals (intermediate, codominant, and dominant trees) using the O-ring test statistic with a fixed adult pattern and heterogeneous Poisson recruit pattern for American beech and sugar maple in the north and south plot. Gray bands indicate complete spatial randomness and black lines represent the observed pattern.

5. Conclusions

The dramatic changes in species composition described here have also been observed in many parts of the Eastern Deciduous Forest region, and may very well lead to collapses in biodiversity and potentially in ecosystem functioning as well (Nuttall et al., 2013). These changes thus pose serious challenges for the management of these systems, and strategies that promote diversity and regeneration of xeric species must be developed to retain the economic, aesthetic, and historic value of these forests. While our results indicate that variation in local species-interactions likely plays a relatively minor role in shifting the composition from oak to beech-maple dominance over a 15-year period, we expect that the use of neighborhood modeling will become an increasingly important tool for both predicting and reversing mesophication trends across the EDF. A particularly important next step will be to better understand the interactive roles between allogenic and autogenic processes in driving shifts in the EDF region, and we expect that neighborhood models will be a valuable tool in this regard due to their ability to incorporate multiple variables and interaction terms. Data from existing plot networks such as the USDA Forest Inventory and Analysis (FIA) Program, and the Smithsonian Forest-GEO network can be parameterized using similar models to the ones presented here for Dysart Woods, but over a much larger geographic area. Furthermore, we show that the incorporation of a simple and easy-to-measure decline index into such

models leads to high prediction ability for individual tree mortality, which could also be incorporated into these larger monitoring efforts. Eventually, we expect that results from such models can provide insight into the primary drivers of mesophication, which can then be directly targeted in management schemes. For example, we predict that selective thinning regimes that promote oak mast seeding and reduce interspecific competition with species such as maple, combined with strategies that reduce disturbances expected to promote mesic species (e.g., deer exclusion and burning), will be necessary to reverse these trends.

Acknowledgements

Funding for this project was provided through two original work grants from the Ohio University Graduate Student Senate, an applied research scholarship from Phi Delta Theta fraternity, and through the Dysart Woods travel fund. We sincerely thank Drs. Glenn Matlack and Jared DeForest for their help improving previous versions of this manuscript.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.foreco.2016.12.032>.

References

- Augsburger, C.K., 1984. Seedling survival of tropical tree species: interactions of dispersal distance, light-gaps, and pathogens. *Ecology* 65, 1705–1712.
- Bagchi, R., Gallery, R.E., Gripenberg, S., Gurr, S.J., Narayan, L., Addis, C.E., Freckleton, R.P., Lewis, O.T., 2014. Pathogens and insect herbivores drive rainforest plant diversity and composition. *Nature* 506, 85–88.
- Baribault, T.W., Kobe, R.K., Rothstein, D.E., 2010. Soil calcium, nitrogen, and water are correlated with aboveground net primary production in northern hardwood forests. *For. Ecol. Manage.* 260, 723–733.
- Barnes, B.V., Zak, D.R., Denton, S.R., Spurr, S.H., 1998. *Forest Ecology*. John Wiley and Sons, New York, NY.
- Buchanan, M.L., Hart, J.L., 2012. Canopy disturbance history of old-growth *Quercus alba* sites in the eastern United States: examination of long-term trends and broad-scale patterns. *For. Ecol. Manage.* 267, 28–39.
- Burns, R.M., Honkala, B.H., 1990. *Silvics of North America: 1. Conifers; 2. Agriculture Handbook 654*. U.S. Department of Agriculture, Forest Service, Washington, DC.
- Canham, C.D., LePage, P.T., Coates, K.D., 2004. A neighborhood analysis of canopy tree competition: effects of shading versus crowding. *Can. J. For. Res.* 34, 778–787.
- Comita, L.S., Muller-Landau, H.C., Aguilar, S., Hubbell, S.P., 2010. Asymmetric density dependence shapes species abundances in a tropical tree community. *Science* 329, 330–332.
- Comita, L.S., Queenborough, S.A., Murphy, S.J., Eck, J.L., Xu, K., Krishnadas, M., Beckman, N., Zhu, Y., 2014. Testing predictions of the Janzen-Connell hypothesis: a meta-analysis of experimental evidence for distance- and density-dependent seed and seedling survival. *J. Ecol.* 102, 845–856.
- Connell, J.H., 1971. On the role of natural enemies in preventing competitive exclusion in some marine animals and in rain forest trees. In: den Boer, P.J., Gradwell, G.R. (Eds.), *Dynamics of Populations*. Centre for Agricultural Publication and Documentation, Wageningen, Netherlands, pp. 298–312.
- Connell, J.H., Tracey, J.G., Webb, L.J., 1984. Compensatory recruitment, growth, and mortality as factors maintaining rain forest tree diversity. *Ecol. Monogr.* 54, 141–164.
- DeForest, J.L., McCarthy, B.C., 2011. Diminished soil quality in an old-growth, mixed mesophytic forest following chronic acid deposition. *Northeast. Nat.* 18, 177–184.
- Dyer, J.M., 2006. Revisiting the deciduous forests of Eastern North America. *Bioscience* 56, 341–352.
- Frazer, G.W., Canham, C.D., Lertzman, K.P., 1999. *Gap Light Analyzer (GLA), Version 2.0: imaging software to extract canopy structure and gap light transmission indices from true-colour fisheye photographs, users manual and program documentation*. 1999. Simon Fraser University, Burnaby, British Columbia, and the Institute of Ecosystem Studies, Millbrook, New York.
- Hubbell, S.P., Ahumada, J.A., Condit, R., Foster, R.B., 2001. Local neighborhood effects on long-term survival of individual trees in a neotropical forest. *Ecol. Res.* 16, 859–875.
- Janzen, D.H., 1970. Herbivores and the number of tree species in tropical forests. *Am. Nat.* 104, 501–528.
- John, R., Dalling, J.W., Harms, K.E., Yavitt, J.B., Stallard, R.F., Mirabello, M., Hubbell, S. P., Valencia, R., Navarrete, H., Vallejo, M., Foster, R.B., 2007. Soil nutrients influence spatial distributions of tropical tree species. *Proc. Natl. Acad. Sci.* 104, 864–869.
- Johnson, D.J., Beaulieu, W.T., Bever, J.D., Clay, K., 2012. Conspecific negative density dependence and forest diversity. *Science* 336, 904–907.
- Kobe, R.K., Vriesendorp, C.F., 2011. Conspecific density dependence in seedlings varies with species shade tolerance in a wet tropical forest. *Ecol. Lett.* 14, 503–510.
- Lafer, N., Wistendahl, W., 1970. Tree composition of Dysart Woods, Belmont County, Ohio. *Castanea* 35, 302–308.
- Labbers, J.H.R., Clark, J.S., Beckage, B., 2002. Density-dependent mortality and the latitudinal gradient in species diversity. *Nature* 417, 732–735.
- Legendre, P., Fortin, M.J., 1989. Spatial pattern and ecological analysis. *Plant Ecol.* 80, 107–138.
- Liebold, A.M., Gurevitch, J., 2002. Integrating the statistical analysis of spatial data in ecology. *Ecography* 25, 553–557.
- Liu, J., Ashton, P.S., 1995. Individual-based simulation models for forest succession and management. *For. Ecol. Manage.* 73, 157–175.
- McCarthy-Neumann, S., Kobe, R.K., 2010. Conspecific and heterospecific plant-soil feedbacks influence survivorship and growth of temperate tree seedlings. *J. Ecol.* 98, 408–418.
- McCarthy, B.C., 1997. Lab protocols for the testing of Eastern Deciduous Forest soils. URL, <https://www.ohio.edu/plantbio/staff/mccarthy/Soils_Lab_Protocols.pdf>.
- McCarthy, B.C., Small, C.J., Rubino, D.L., 2001. Composition, structure and dynamics of Dysart Woods, an old-growth mixed mesophytic forest of southeastern Ohio. *For. Ecol. Manage.* 140, 193–213.
- McEwan, R.W., Dyer, J.M., Pederson, N., 2011. Multiple interacting ecosystem drivers: toward an encompassing hypothesis of oak forest dynamics across eastern North America. *Ecography* 34, 244–256.
- McLaughlin, D., Gizyn, W., McIver, W., Kinch, C., 1992. A quantitative tree crown rating system for deciduous forest health surveys, some results from Ontario. In: Manion, P.D., Lachance, D. (Eds.), *Forest Decline Concepts*. The American Phytopathological Society, St. Paul, MN, pp. 137–154.
- McShea, W.J., Healy, W.M., Devers, P., Fearer, T., Koch, F.H., Stauffer, D., Waldon, J., 2007. Forestry matters: decline of oaks will impact wildlife in hardwood forests. *J. Wildl. Manage.* 71, 1717–1728.
- Mehlich, A., 1984. Mehlich 3 soil test extractant: a modification of Mehlich 2 extractant. *Commun. Soil Sci. Plant Anal.* 15, 1409–1416.
- Nowacki, G., Abrams, M., 2008. The demise of fire and “mesophication” of forests in the Eastern United States. *Bioscience* 58, 123–138.
- Nuttall, T., Royo, A.A., Adams, M.B., Carson, W.P., 2013. Historic disturbance regimes promote tree diversity only under low browsing regimes in eastern deciduous forest. *Ecol. Monogr.* 83, 3–17.
- Oliver, C.D., Larson, B.C., 1996. *Forest Stand Dynamics*. McGraw Hill, New York, NY.
- Richards, J.D., Hart, J.L., 2011. Canopy gap dynamics and development patterns in secondary *Quercus* stands on the Cumberland Plateau, Alabama, USA. *For. Ecol. Manage.* 262, 2229–2239.
- Rubel, N., Jenny, P.C., Plunkett, M.K., 1981. *Soil Survey of Belmont County*. USDA, Soil Conservation Service, Washington, DC, Ohio.
- Sheil, D., Burslem, D.F.R.P., Alder, D., 1995. The interpretation and misinterpretation of mortality rate measures. *J. Ecol.* 83, 331–333.
- Tansley, A.G., 1935. The use and abuse of vegetational concepts and terms. *Ecology* 16, 284–307.
- Whitney, G.G., 1994. *From coastal wilderness to fruited plain: a history of environmental change in temperate North America from 1500 to the present*. Cambridge University Press, New York, NY.
- Wiegand, T., Martínez, I., Huth, A., 2009. Recruitment in tropical tree species: revealing complex spatial patterns. *Am. Nat.* 174, E106–140.
- Wiegand, T., Moloney, K.A., 2004. Rings, circles, and null-models for point pattern analysis in ecology. *Oikos* 104, 209–229.
- Wiegand, T., Moloney, K.A., 2014. *Handbook of Spatial Point-Pattern Analysis in Ecology*. CRC Press, Boca Raton, FL, USA.
- Zhao, D., Borders, B., Wang, M., Kane, M., 2007. Modeling mortality of second-rotation loblolly pine plantations in the Piedmont/Upper Coastal Plain and Lower Coastal Plain of the southern United States. *For. Ecol. Manage.* 252, 132–143.
- Zhao, D., Borders, B., Wilson, M., 2004. Individual-tree diameter growth and mortality models for bottomland mixed-species hardwood stands in the lower Mississippi alluvial valley. *For. Ecol. Manage.* 199, 307–322.
- Zhu, Y., Comita, L.S., Hubbell, S.P., Ma, K., 2015. Conspecific and phylogenetic density-dependent survival differs across life stages in a tropical forest. *J. Ecol.* 103, 957–966.